

tsvc_c2/k53 · k53.c · full_nospeed_v4 · claude-haiku-4-5-20251001 · BROKEN

```
-2 / +10 lines of 13      · 1 parallel region(s) added      · 3 of 7 candidate(s) kept      · RESTRUCTURING (9 code line(s) added, 2 removed)      · restructuring level 0
```

BEFORE

```
1  #include "data.h"
```

```
3 real t kernel k53(void)
```

4 {

```
5     for (int nl = 0; nl < iterations; nl++) {
```

```
6         for (long i = 1; i < LEN 1D; i++) {
```

```
7         u[ju[i]] += v[kv[i]] * c[i];
```

```
8         v[jv[i]] = u[ku[i]] * d[i] + c[i];
```

9 }

```
10 dummy(a, b, c, d, e);
```

```
12     return (real t)0;
```

13 }

AFTER

```
1  #include "data.h"
```

```
3 real t kernel k53(void)
```

4 {

```
5     for (int nl = 0; nl < iterations; nl++) {
```

```
6      real t v work[LEN 1D]
```

```
7         for (long k = 0; k < LEN_1D; k++) {
```

```
8         v_work[k] = v[k];
```

```
10     for (long i = 1; i < LEN 1D; i++) {
```

```
l1      u[ju[i]] += v_work[kv[i]] * c[i];
```

```
12         v_work[jv[i]] = u[ku[i]] * d[i] + c[i];
```

```
14      #pragma omp parallel for firstprivate(v_work)
```

```
15     for (long k = 0; k < LEN_1D; k++) {
```

```
16         v[k] = v_work[k];
```

17

```
18 dummy(a, b, c, d, e);
```

19 }

```
20     return (real t)0;
```

21 }

file